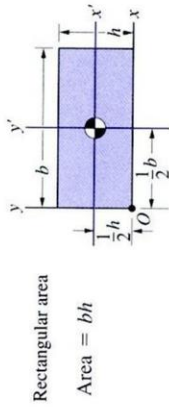


moment of inertia about the x axis I_x
 moment of inertia about the y axis I_y
 product of inertia I_{xy}

$$I_x = \int_A y^2 dA, \quad I_y = \int_A x^2 dA, \quad I_{xy} = \int_A xy dA$$

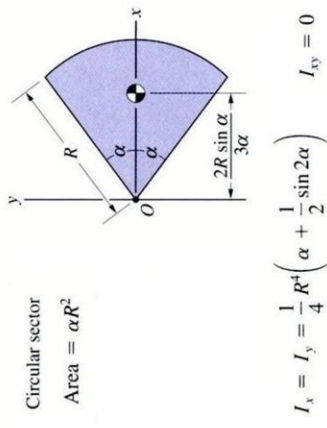
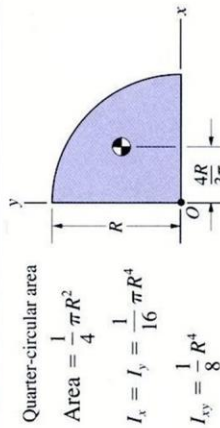
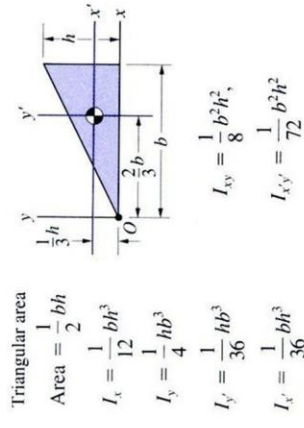
polar moment of inertia about O is

$$J_O = \int_A r^2 dA = \int_A (x^2 + y^2) dA = I_x + I_y$$

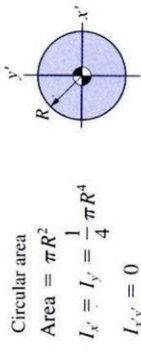
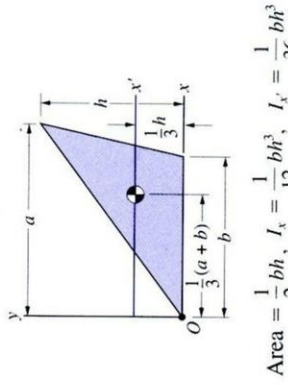


$$I_x = \frac{1}{3}bh^3, \quad I_y = \frac{1}{3}hb^3, \quad I_{xy} = \frac{1}{4}b^2h^2$$

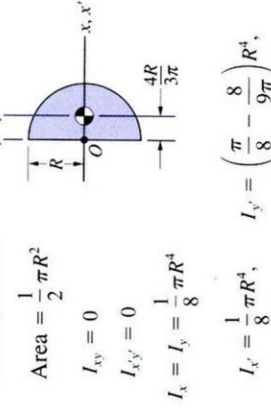
$$I_x' = \frac{1}{12}bh^3, \quad I_y' = \frac{1}{12}hb^3, \quad I_{xy}' = 0$$



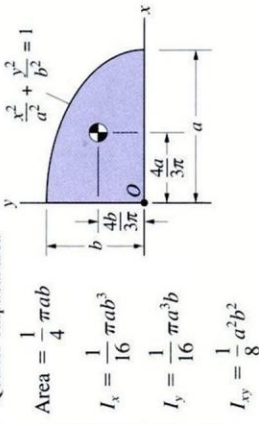
Triangular area



Semicircular area



Quarter-elliptical area



Spandrel

